Inequalities — problem 7.37

A. Engel — Problem-Solving Strategies · 1998

If $x_i > 0$ and $x_i y_i - z_i^2 > 0$ for $i \leq n$, then

$$\frac{n^3}{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right) - \left(\sum_{i=1}^n z_i\right)^2} \leq \sum_{i=1}^n \frac{1}{x_i y_i - z_i^2}.$$

Prove this inequality for $n = 2$ (IMO 1969), and then also generally.